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### Air intake plenum with struts

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#### Abstract

An air intake plenum for a gas turbine engine is provided that includes an axial centerline, first and second gas path surfaces, and a plurality of struts. The gas path surfaces are spaced apart from one another and define at least a portion of a plenum interior. The struts extend lengthwise between the gas path surfaces. Each strut has a cross-section geometry. The cross-section geometry has a center, and major and minor axes. The struts are circumferentially spaced apart from one another within the plenum. Each strut is oriented at a clocking angle  $\theta$ . The clocking angle  $\theta$  for each respective strut is disposed between the major axis of that strut and a line that intersects the cross-section geometry center of that strut and the axial centerline. At least one strut of the plurality of struts is oriented at a clocking angle  $\theta$  that is greater than zero.

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## Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/723,012 filed Apr. 18, 2022 which is hereby incorporated herein by reference in its entirety.

### TECHNICAL FIELD

(1) The present disclosure relates to aircraft intake air plenums in general, and to aircraft intake air plenums having struts in particular.

### BACKGROUND INFORMATION

(2) In some aircraft applications, a compressor section for a turboshaft gas turbine engine application may receive intake air from an air intake plenum that extends completely around (e.g., 360°) a radial compressor intake. Intake air from aircraft ducting enters an inlet opening in the air intake plenum and is distributed circumferentially around the plenum. The intake air within the plenum is turned into the radial intake of the compressor section. In some embodiments, airflow exiting the air intake plenum encounters a plurality of inlet guide vanes (IGVs) prior to entering a

first compressor stage, which IGVs function to apply a desired orientation of the airflow entering the first compressor stage.

(3) Most plenum designs include a plurality of struts that extend through the interior of the plenum between gas path walls that define the interior of the plenum. The struts are stationary and are circumferentially spaced apart to provide structural support to the plenum/intake. It is known to dispose the struts in a uniform spacing around the circumference of the plenum; e.g., a plenum having four (4) struts may have a strut positioned every ninety degrees ( $90^\circ$ ) around the entire three-hundred and sixty degree ( $360^\circ$ ) circumference, etc. The exact number of struts in a plenum/intake may differ in different plenum designs.

(4) Each strut has a lengthwise axis that extends between gas path surfaces, and each has a cross-section shape (i.e., cross-section perpendicular to the lengthwise axis). Struts can have a variety of different cross-section geometries although most have an oval/elliptical geometry defined by a major axis extending between vertexes and a minor axis perpendicular to the major axis (major axis > minor axis). In some instances, a strut may have a cross-section geometry that is airfoil-like. In these embodiments, the chord of the airfoil may be equated to a "major axis".

(5) Struts disposed within an intake plenum almost always create an impediment to airflow traveling within the plenum. Airflow encountering a strut typically must pass by that strut. Each strut therefore typically has a stagnation point where the local velocity of the airflow is zero and the airflow separates to pass around the strut. A consequence of the separated airflow is a "pressure loss" or "low momentum" region (e.g., low pressure, low velocity, etc.) disposed on the side of the strut opposite the side of airflow impingement. These loss regions are indicative of undesirable energy or momentum losses within the airflow that are detrimental to airflow within the plenum.

(6) It is often incorrectly assumed that intake air enters the inlet opening of the engine plenum in a uniform manner; e.g., uniform in fluid flow parameters such as volumetric flow rate, velocity, pressure, etc. In most instances, however, intake air enters the inlet opening of the engine plenum in a non-uniform manner; e.g., the fluid flow parameters of the intake air varies across the plenum inlet as a function of position. In fact, it is often the case that ducting that supplies the plenum is shortened for reasons like improving the external appearance for aircraft or reducing the cross-sectional area. Such modifications often require very compact ducting upstream of the plenum and the compact ducting often produces non-uniform flow at the engine air intake plenum inlet with consequential impact on compressor performance.

(7) What is needed is an improved air intake plenum strut arrangement that decreases airflow losses within the plenum, and one that accommodates non-uniform airflow into the plenum.

## SUMMARY

(8) According to an aspect of the present disclosure, an air intake plenum for a gas turbine engine is provided that includes an axial centerline, a first gas path surface, a second gas path surface, and a plurality of struts. The second gas path surface is spaced apart from the first gas path surface. The first and second gas path surfaces define at least a portion of a plenum interior configured for air passage. The plurality of struts extend lengthwise between the first and second gas path surfaces, through the plenum interior. Each strut has a cross-section geometry within a plane perpendicular to a lengthwise axis of the strut. The cross-section geometry having a center, a major axis and a minor axis, where the major axis is greater than the minor axis. The plurality of struts are disposed circumferentially spaced apart from one another within the plenum. Each of the plurality of struts is oriented at a clocking angle  $\theta$ . The clocking angle  $\theta$  for each respective strut is disposed between the major axis of that respective strut and a line that intersects the cross-section geometry center of that strut and the axial centerline. At least one strut of the plurality of struts is oriented at a clocking angle  $\theta$  that is greater than zero.

(9) In any of the aspects or embodiments described above and herein, the air intake plenum may be configured to provide air to a compressor section of a gas turbine engine.

(10) In any of the aspects or embodiments described above and herein, the at least one strut of the

plurality of struts may include a first strut oriented at a clocking angle  $\theta_1$  that is greater than zero and a second strut oriented at a clocking angle  $\theta_2$  that is greater than zero, wherein  $\theta_2$  is greater than  $\theta_1$ .

(11) In any of the aspects or embodiments described above and herein, the at least one strut of the plurality of struts is oriented at a clocking angle  $\theta$  equal to zero.

(12) In any of the aspects or embodiments described above and herein, each strut of the plurality of struts may be circumferentially positioned at a respective circumferential angle  $\phi$ , and the clocking angle  $\theta$  of a given strut within the plurality of struts may be a function of the circumferential angle  $\phi$  of that particular strut.

(13) In any of the aspects or embodiments described above and herein, the plurality of struts may be uniformly circumferentially spaced within the plenum.

(14) In any of the aspects or embodiments described above and herein, the plurality of struts may be non-uniformly circumferentially spaced within the plenum.

(15) In any of the aspects or embodiments described above and herein, the air intake plenum may include an air inlet, and the majority of the struts may be disposed within the plenum in a portion of the plenum disposed on an opposite side of the axial centerline than the air inlet.

(16) In any of the aspects or embodiments described above and herein, the at least one strut of the plurality of struts may have a symmetrical cross-section geometry.

(17) In any of the aspects or embodiments described above and herein, the cross-section geometry center of the at least one strut of the plurality of struts may be disposed at an intersection of the major axis and the minor axis.

(18) In any of the aspects or embodiments described above and herein, the at least one strut of the plurality of struts may include at least one first strut having a first cross-section geometry and at least one second strut having a second cross-section geometry, wherein the first cross-section geometry is different from the second cross-section geometry.

(19) In any of the aspects or embodiments described above and herein, the plurality of struts includes “N” number of struts, where “N” is an integer greater than two, and each respective strut may be circumferentially positioned at a respective circumferential angle  $\phi$ , and the clocking angle  $\theta$  of that strut is a function of at least one of the circumferential angle  $\phi$  of that particular strut, the cross-section of that particular strut, and the N number of struts.

(20) According to another aspect of the present disclosure, a gas turbine engine is provided that includes an axial centerline, at least one compressor rotor stage having a first bladed rotor rotatable about the axial centerline, at least one turbine rotor stage having a second bladed rotor rotatable about the axial centerline, and an air intake plenum in communication with the at least one compressor rotor stage. The air intake plenum includes a first gas path surface and a second gas path surface, spaced apart from the first gas path surface. The first and second gas path surfaces define at least a portion of a plenum interior configured for air passage. A plurality of struts extend lengthwise between the first and second gas path surfaces through the plenum interior. Each strut has a cross-section geometry within a plane perpendicular to a lengthwise axis of the strut. The cross-section geometry has a center, a major axis and a minor axis, where the major axis is greater than the minor axis. The plurality of struts are disposed circumferentially spaced apart from one another within the plenum. Each of the plurality of struts is oriented at a clocking angle  $\theta$ , wherein the clocking angle  $\theta$  for each respective strut is disposed between the major axis of that respective strut and a line that intersects the cross-section geometry center of that strut and the axial centerline. At least one strut of the plurality of struts is oriented at a clocking angle  $\theta$  that is greater than zero.

(21) According to an aspect of the present disclosure, a method of orienting a plurality of struts within an air intake plenum in a gas turbine engine is provided. The method includes: a) providing an air intake plenum that includes an axial centerline, a first gas path surface, a second gas path surface, and “N” number of struts, where N is an integer equal to or greater than two, and wherein

the second gas path surface is spaced apart from the first gas path surface, wherein the first gas path surface and the second gas path surface define at least a portion of a plenum interior configured for air passage, and the N number of struts extend lengthwise between the first gas path surface and the second gas path surface through the plenum interior, and wherein each strut has a cross-section geometry within a plane perpendicular to a lengthwise axis of the respective strut, the cross-section geometry having a center, a major axis and a minor axis, where the major axis is greater than the minor axis, and wherein each respective strut of the N number of struts is circumferentially positioned at a respective circumferential angle  $\phi$ , and wherein each of the N number of struts is oriented at a clocking angle  $\theta$ , wherein the clocking angle  $\theta$  for each respective strut is disposed between the major axis of that respective strut and a line that intersects the cross-section geometry center of that strut and the axial centerline; and b) determining a clocking angle for each strut of the N number of struts based on at least one of the circumferential angle  $\phi$  of that particular strut, the cross-section of that particular strut, and the N number of struts, and wherein at least one strut of the N number of struts is oriented at a clocking angle  $\theta$  that is greater than zero.

(22) In any of the aspects or embodiments described above and herein, the step of determining a clocking angle may include determining the clocking angle  $\theta$  for each strut as a function of the circumferential angle  $\phi$  for that strut.

(23) In any of the aspects or embodiments described above and herein, a polynomial equation may be used to determine the clocking angle  $\theta$  for each strut as a function of the circumferential angle  $\phi$  for that strut.

(24) In any of the aspects or embodiments described above and herein, the polynomial equation

$$(25) \quad = V^4 - W^3 + X^2 + Y + Z$$

may be used to determine the clocking angle  $\theta$  for each strut as a function of the circumferential angle  $\phi$  for that strut.

(26) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, the following description and drawings are intended to be exemplary in nature and non-limiting.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an example gas turbine engine.

(2) FIG. 2 is an enlarged portion of the gas turbine engine shown in FIG. 1, now including an air intake plenum strut.

(3) FIG. 3 is a diagrammatic sectional view (see Section A-A in FIG. 1) of an air intake plenum embodiment.

(4) FIG. 4 is a diagrammatic view of a strut cross-section geometry.

(5) FIG. 5 is an exemplary fluid flow diagram for an air intake plenum like that shown in FIG. 3.

(6) FIG. 6 is an exemplary entropy diagram for an air intake plenum like that shown in FIG. 3.

(7) FIG. 7 is a diagrammatic sectional view of an air intake plenum embodiment, illustrating struts having respective major axes disposed at particular clocking angles.

(8) FIG. 7A is an enlarged view of the strut 42 also labeled as “strut A” in FIG. 7.

(9) FIG. 8 is a graph of strut clocking angle  $\theta$  (“ $\theta$ ”) versus strut circumferential angle  $\phi$  (“ $\phi$ ”), including a polynomial curve and offset curves.

(10) FIG. 9 is a diagrammatic sectional view of an air intake plenum embodiment, illustrating struts having respective major axes disposed at particular clocking angles, with the struts disposed in a

non-uniform circumferential arrangement.

## DETAILED DESCRIPTION

(11) FIG. 1 diagrammatically illustrates a cross-section of gas turbine engine **20** that includes an air intake plenum **22** in fluid communication with airflow from upstream ducting (not shown), a compressor section **24**, a combustor section **26**, and a turbine section **28** in serial flow communication along an engine axial centerline **30**. The terms “forward” and “aft” are used herein to refer to relative axial positions; e.g., in gas turbine engine **20** example shown in FIG. 1, the compressor section **24** is disposed forward of the turbine section **28** and therefore the turbine section **28** is aft of the compressor section **24**. The combustor section **24** is shown as having at least one rotor stage **32** having a rotor with a plurality of blades disposed around the circumference of the rotor. The turbine section **28** is shown as having a high pressure rotor stage **34** and a low pressure rotor stage **36**, each stage **34**, **36** having a rotor with a plurality of blades disposed around the circumference of the respective rotor. The turbine high pressure rotor stage **34** is in communication with a first spool **38** that is in communication with the compressor rotor stage **32**. The turbine low pressure rotor stage **36** is in communication with a second spool **40** that is in co-axial with the first spool **38**. The second spool **40** may be in communication directly or indirectly (e.g., through a gearbox) with a fan section, a propeller, or the like (not shown). In this example, intake air from aircraft ducting (not shown) enters the air intake plenum **22**, and passes into the compressor section **24**. The compressor section **24** does work on the intake air causing it to increase in pressure and temperature. Within the combustor section **26**, fuel is added to the worked air and the mixture is ignited. Byproducts from the combustion and some amount of worked air (collectively now referred to as “core gas”) subsequently passes into the turbine section **28** where the core gas powers the spools **38**, **40**. The core gas subsequently passes through a turbine exhaust case (which may also include struts) before exiting the gas turbine engine **20**.

(12) The partial section shown in FIG. 2 illustrates an example of where a strut **42** may be located within a compressor air intake plenum **22** and the present disclosure is not limited thereto. Moreover, the present disclosure is not limited to any particular air intake plenum **22** configuration, other than one having a strut **42** configuration as described herein. The intake air plenum **22** may be described as being defined by a first gas path wall **58** and an opposing second gas path wall **60** with a plenum interior **61** disposed there between. Depending on the orientation of the plenum **22**, the first gas path wall **58** may be disposed forward of the second gas path wall **60**, or the first gas path wall **58** may be disposed radially outside of the second gas path wall **60**, or any combination thereof. The strut **42** shown in FIG. 2 extends between gas path walls **58**, **60** along a lengthwise axis **56**.

(13) FIG. 3 diagrammatically illustrates a sectioned air intake plenum **22** with an air inlet **44** and five (5) struts **42** uniformly spaced from one another around the circumference of the plenum **22**; e.g., each strut **42** is positioned about seventy-two degrees ( $72^\circ$ ) or  $2\pi$  radians/5 from adjacent struts **42** around the plenum **22** circumference. A plurality of inlet guide vanes (IGVs) **46** are disposed radially inside of the struts **42** and around a compressor inlet **48** (the IGVs **46** may not be axially aligned with the struts **42**). The struts **42** all have the same configuration; e.g., an elliptical shape having a major axis **50** extending through a first vertex **52** and a second vertex **54** and a minor axis **55** (see FIG. 4). The major axis **50** is perpendicular to the minor axis **55**. In the embodiment shown in FIG. 3, the major axis **50** of each strut **42** is collinear with a respective line that intersects the axial centerline **30** of the compressor section **24**. In this diagram, the air flow entering the compressor air intake plenum **22** via the air inlet **44** is shown diagrammatically as being uniform across the inlet **44**; e.g., the fluid flow parameters of the inlet air are uniform across the inlet **44**.

(14) FIG. 5 is an exemplary fluid flow diagram for an air intake plenum **22** like that shown in FIG. 3. The relative airflow velocities are shown diagrammatically as a function of absolute Mach number. Intake air entering the plenum **22** travels circumferentially both clockwise and

counterclockwise around the axial centerline **30**. The relative differences in airflow velocity are indicated by different region numbers (R1, R2, R3) with each region depicted by different cross-hatching; i.e., airflow velocity indicated by region R1 is greater than the airflow velocity indicated by region R2, and the airflow velocity indicated by region R2 is greater than the airflow velocity indicated by region R3. It should be noted that the airflow velocity regions R1-R3 in FIG. 5 are simplified representations to facilitate illustration of the same. Nonetheless, the aforesaid airflow velocity regions accurately depict an example of circumferentially non-uniform airflow velocities within a plenum **22**. The fluid flow stagnation point **43** for each strut **42** is shown; i.e., the point where the local airflow velocity is zero and the airflow separates to pass around the strut **42**. FIG. 5 clearly shows a) the airflow velocity is circumferentially non-uniform around the plenum **22**; and b) the presence of loss regions (e.g., flow regions having low pressure, low velocity—"low momentum") disposed relative to respective struts **42**. FIG. 6 illustrates an exemplary entropy diagram for an air intake plenum **22** like that shown in FIG. 3. Here again, loss regions disposed relative to respective struts **42** are clearly visible. FIGS. 5 and 6 also diagrammatically show how airflow characteristics (e.g., strut wakes/loss regions) can extend into downstream IGVs **46**. Strut wakes can cause high cycle fatigue and impact vibration on a downstream rotor and can lead to structural damage in rotors and/or IGVs.

(15) Aspects of the present disclosure include an air intake plenum **22** with a plurality of struts **42** disposed around the circumference of the plenum **22**. As stated above, each strut **42** may extend along a lengthwise axis **56** between a first gas path surface **58** and a second gas path surface **60** and the gas path surfaces **58**, **60** define at least a portion of a plenum interior **61**; e.g., see FIG. 2. The air intake plenum **22** includes an air inlet **44** that is connected to a source of air; e.g., a source of air directed to the plenum air inlet **4** via ducting. In FIG. 7, the air inlet **44** is shown as a single inlet disposed on an upper portion of the plenum **22**. The present disclosure is not limited to any particular air inlet **44** positioning relative to the plenum **22**, any particular air inlet **44** configuration, and is not limited to a single air inlet **44**.

(16) The air intake plenum **22** shown in FIG. 7 depicts five (5) struts **42** disposed circumferentially around a compressor inlet **48**. A plurality of IGVs **46** are disposed radially between the struts **42** and compressor inlet **48** (the IGVs **46** may not be axially aligned with the struts **42**). The present disclosure can be used with fewer or more than five (5) struts **42** and may be used in embodiments where IGVs **46** are used and in embodiments where IGVs **46** are not used.

(17) Each of the struts **42** shown in FIG. 7 has a cross-section geometry that reflects the strut **42** geometry in a plane perpendicular to the lengthwise axis **56** (see FIG. 2) of the strut **42**. FIG. 7A is an enlarged view of the strut **42** also labeled as "strut A" in FIG. 7. Each strut cross-section geometry includes a major axis **50** and a minor axis **55**, where the major axis **50** is greater than the minor axis **55**, and the major and minor axes **50**, **55** are perpendicular one another. Non-limited examples of strut cross-section geometry include symmetrical geometries such as an ellipse or a non-circular oval, or a symmetrical airfoil, or the like, or non-symmetrical geometries such as certain non-symmetrical airfoil geometries, or the like. In terms of airfoil geometries, for purposes of description herein the chord of the airfoil geometry may be equated to the major axis **50**. The present disclosure is not limited to any particular strut **42** cross-section geometry other than one having a major axis **50** that is greater than the minor axis **55**. The struts **42** within some embodiments of the present disclosure may all have the same cross-section geometry. The struts **42** within some embodiments of the present disclosure may include at least one strut **42** having a first cross-section geometry and at least one strut having a second cross-section geometry, wherein the first and second cross-section geometries are not the same; i.e., in some embodiments not all of the struts **42** within the plenum **22** may have the same cross-section geometry.

(18) The circumferential position of the struts **42** within the air intake plenum **22** may be described in terms of radians to facilitate the description. The angular position of each strut **42** within the plenum circumference is indicated by a circumferential angle phi (" $\phi$ "). For example, since two pi

( $2\pi$ ) radians=360 degrees, a strut **42** located at top dead center may be described as being positioned at an angle phi equal to zero ( $\phi=0$ ) and a strut **42** located at bottom dead center may be described as being positioned at an angle phi equal to pi ( $\phi=\pi$ ). FIG. 7 illustrates the five struts **42** uniformly positioned around the circumference of the plenum **22**; e.g., each strut **42** is equidistant to either adjacent strut **42**. To be clear, the present disclosure is not limited to any particular number of struts **42** and is not limited to struts **42** that are uniformly positioned around the circumference of the plenum **22**.

(19) Aspects of the present disclosure reflect the discovery that the orientation of the major axis **50** of each strut **42** as a function of the respective strut's circumferential position can yield significant improvements in airflow within the air intake plenum **22**; e.g., a significant decrease in airflow losses associated with respective struts **42**, etc. It is known to orient struts **42** within an air plenum **22** so that the major axis **50** of each strut **42** is collinear with a respective line that intersects the axial centerline **30** of the compressor section **24**, which centerline **30** is typically the axial/rotational centerline **30** of the gas turbine engine **20**; e.g., see FIG. 3. According to aspects of the present disclosure, the major axis **50** of a strut **42** may be skewed by an angle theta ( $\theta$ )—which may be referred to herein as a “clocking angle”) from a line **62** that intersects both the cross-section geometry center **64** of the strut **42** and the axial centerline **30** of the compressor section **24**. The degree to which the major axis **50** of a strut **42** is skewed by an angle theta ( $\theta$ ) according to the present disclosure may be determined, as described herein, by factors including the number of struts **42** within the plenum **22**, the geometry of the struts **42**, the circumferential position of a strut **42** within the plenum **22**, flow characteristics of the inlet air entering the plenum **22** (e.g., the degree of uniformity or non-uniformity, velocity, etc.), the geometry of the plenum **22**, and the like. To be clear, and as will be described herein, different struts **42** may have their respective major axis **50** skewed at different angle thetas (e.g., a first strut having a major axis disposed at a first theta angle, a second strut having a major axis disposed at a second theta angle that is different from the first theta angle, etc.) and certain struts **42** may have their respective major axis **50** skewed at an angle theta that is the same as one or more other struts **42** (e.g., a third and fourth strut each having a major axis **50** disposed at a given theta angle).

(20) For struts **42** having a symmetrical cross-section geometry (e.g., an ellipse, an oval, etc.), the “center **64**” of the strut **42** may be the center of that cross-section geometry. For example, the center of an elliptical cross-section geometry is the intersection point between the major and minor axes of the ellipse. For struts **42** having a non-symmetrical cross-section geometry (e.g., some airfoils, etc.), the “center **64**” of the strut **42** may be chosen and that center point **64** may be used for each such strut **42** having that non-symmetrical cross-section geometry.

(21) Orienting one or more struts **42** within an air plenum **22** so that the major axis **50** of a particular strut **42** is skewed by an angle theta (as described above) permits each strut **42** to be oriented for improved alignment of that strut **42** with airflow local to that strut **42** within the plenum **22**. The resulting strut **42** orientations produce less loss within the plenum **22**, improved airflow characteristics into the compressor inlet **48**, and consequent improved engine performance with likely no gain in weight and/or increase in manufacturing cost.

(22) Referring to FIG. 7, to facilitate the description herein the struts **42** disposed within the air intake plenum **22** are labeled as A, B, C, D, E in relative clockwise position around the circumference of the plenum **22**. In this non-limiting example, the struts **42** are uniformly spaced from one another around the circumference of the plenum **22**. Strut “A” is located at top dead center ( $\phi=0$ ) and is oriented to have its major axis **50** at a clocking angle theta1 ( $\theta_1$ ) equal to zero ( $\theta_1=0$ ), meaning that the major axis **50** of strut “A” is collinear with a line **62A** that intersects the cross-section geometry center **64** of strut “A” and the axial centerline **30** of the compressor section. Strut “B” is located at a circumferential angle phi2 ( $\phi_2=X$  radians, wherein “X” is  $2\pi$  radians/5 displaced from top dead center and is oriented to have its major axis **50** at a clocking angle theta2 ( $\theta_2$ ), meaning that the major axis **50** of strut “B” is skewed from a line **62B** that intersects the



cross-section geometry center **64** of strut “B” and the axial centerline **30** of the compressor section **24** by a clocking angle equal to  $\theta_2$  (“ $\theta_2$ ”). Strut “C” is located at a circumferential angle  $\phi_3$  ( $\phi_3=2\pi$  radians) displaced from top dead center and is oriented to have its major axis **50** at a clocking angle equal to  $\theta_3$  (“ $\theta_3$ ”), meaning that the major axis **50** of strut “C” is skewed from a line **62C** that intersects the cross-section geometry center of strut “C” and the axial centerline **30** of the compressor section **24** by a clocking angle  $\theta_3$  (“ $\theta_3$ ”). Strut “D” is located at a circumferential angle  $\phi_4$  ( $\phi_4=3\pi$  radians) displaced from top dead center and is oriented to have its major axis **50** collinear with a line **62D** that intersects the cross-section geometry center **64** of strut “D” and the axial centerline **30** of the compressor section **24**, meaning that the major axis **50** of strut “D” is disposed at a clocking angle  $\theta_4$  equal to zero ( $\theta_4=0$ ). Strut “E” is located at a circumferential angle  $\phi_5$  ( $\phi_5=4\pi$  radians) displaced from top dead center and is oriented to have its major axis **50** at a clocking angle  $\theta_5$  (“ $\theta_5$ ”), meaning that the major axis **50** of strut “E” is skewed from a line **62E** that intersects the cross-section geometry center **64** of strut “E” and the axial centerline **30** of the compressor section **24** by a clocking angle  $\theta_5$  (“ $\theta_5$ ”). It can be seen from the description of the example given above and FIG. 7 that the clocking angles ( $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5$ ) of the respective struts may vary from one another or some may equal one another; e.g., in the example above,  $\theta_1=\theta_4=0$ . The above described strut arrangement (e.g., circumferential angles  $\phi_1$ - $\phi_5$  and respective clocking angles  $\theta_1$ - $\theta_5$ ) is provided herein to illustrate the present disclosure and the present disclosure is not limited thereto.

(23) The particular angle  $\theta$  (“ $\theta$ ”) for each strut **42** may be analytically determined, for example using fluid flow modeling software that may be tailored according to the present disclosure to consider factors such as the number of struts **42** within the plenum **22**, the geometry of the struts **42**, the circumferential position of a strut **42** within the plenum **22**, flow characteristics of the inlet air entering the plenum **22** (e.g., the degree of uniformity or non-uniformity, velocity, etc.), the geometry of the plenum **22**, and the like. In some embodiments, the skew angle  $\theta$  (“ $\theta$ ”) for each strut **42** may be characterized by a polynomial equation such as Equation 1 below:

$$(24) \quad \theta = V^4 - W^3 + X^2 + Y + Z \quad (\text{Eqn. 1})$$

where the coefficients V, W, X, Y, and Z may be analytically determined as stated above using fluid flow modeling software or other applicable method. FIG. 8 graphically illustrates the relationship between the clocking angle  $\theta$  (“ $\theta$ ”) and the angular position  $\phi$  (“ $\phi$ ”) of a strut **42** within the plenum **22** circumference. Hence, a mathematical expression such as, but not limited to, Equation 1, that accounts for relevant variables can be used as a tool to determine an acceptable clocking angle  $\theta$  (“ $\theta$ ”) for a given strut **42**. The graphed function based on polynomial Equation 1 is further useful to consider plus and/or minus offsets in the clocking angle  $\theta$  (“ $\theta$ ”); e.g., a ten degree offset ( $\pm 10^\circ$ ) in the variable  $\theta$  (“ $\theta$ ”). It should be noted that the example clocking angle orientations shown in FIG. 7, Equation 1, and the graph of FIG. 8 are based on a chosen engine axial rotation direction (i.e., counterclockwise—as indicated in FIG. 7). If the axial rotation of the compressor rotor stage **32** is opposite (i.e., clockwise) the convention for circumferential angle  $\phi$  (“ $\phi$ ”) would be reversed.

(25) In the above examples, the struts **42** are shown and described as being uniformly spaced from one another around the circumference of the plenum **22**. The present disclosure is not limited to an air intake plenum **22** with circumferentially uniformly spaced struts **42**. According to some aspects of the present disclosure, an air intake plenum **22** may include a plurality of struts **42** within an intake plenum **22** that are non-uniformly spaced around the circumference of the intake plenum **22**. The embodiment shown in FIG. 9 illustrates a strut **42** arrangement wherein the plurality of struts **42** are disposed closer to the end **67** of the plenum **22** opposite the air inlet **44** of the plenum **22**; e.g., a majority of the struts **42** are disposed beneath a horizontal axis **66** line extending through the rotational axis **30** of the compressor section **24**. In the particular embodiment shown in FIG. 9 four of five struts **42** are disposed below the aforesaid horizontal axis line **66**. Since airflow velocity

decreases from the plenum air inlet **44** to the opposite end **67** of the plenum **22**, positioning struts **42** in a non-uniform arrangement as shown in FIG. **9** (e.g., spaced away from the air inlet **44**) may, by itself, yield advantageous airflow characteristics within the plenum **22**. In some embodiments (as shown in FIG. **9**), a non-uniform strut **42** arrangement may be coupled with struts **42** disposed at clocking angle  $\theta$  (06, 07, 08, 09, 010) as described above to yield still further improvements in airflow characteristics within an air intake plenum **22**. A desirable clocking angle  $\theta$  for each strut **42** in such an arrangement may be determined as described above; e.g., using a mathematical expression like Equation 1.

(26) While the principles of the disclosure have been described above in connection with specific apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the disclosure. Specific details are given in the above description to provide a thorough understanding of the embodiments. However, it is understood that the embodiments may be practiced without these specific details.

(27) The singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. For example, the term “comprising a specimen” includes single or plural specimens and is considered equivalent to the phrase “comprising at least one specimen.” The term “or” refers to a single element of stated alternative elements or a combination of two or more elements unless the context clearly indicates otherwise. As used herein, “comprises” means “includes.” Thus, “comprising A or B,” means “including A or B, or A and B,” without excluding additional elements.

(28) It is noted that various connections are set forth between elements in the present description and drawings (the contents of which are included in this disclosure by way of reference). It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. Any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option.

(29) No element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprise”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(30) While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the disclosures—such as alternative materials, structures, configurations, methods, devices, and components, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein. For example, in the exemplary embodiments described above within the Detailed Description portion of the present specification, elements may be described as individual units and shown as independent of one another to facilitate the description. In alternative embodiments, such elements may be configured as combined elements.

## Claims

1. An air intake plenum for a gas turbine engine, comprising: an axial centerline; a first gas path surface; a second gas path surface, spaced apart from the first gas path surface, wherein the first gas path surface and the second gas path surface define at least a portion of a plenum interior configured for air passage; and a plurality of struts that extend lengthwise between the first gas path surface and the second gas path surface through the plenum interior, each strut having a cross-section geometry within a plane perpendicular to a lengthwise axis of the strut, the cross-section geometry having a cross-section geometry center, a major axis, and a minor axis, where the major axis is greater than the minor axis, wherein the cross-section geometry is the same for each strut, and the cross-section geometry center disposed at an intersection of the major axis and the minor axis, and the cross-section is symmetrical about the cross-section geometry center; wherein the plurality of struts are disposed circumferentially spaced apart from one another within the air intake plenum, and the plurality of struts are disposed circumferentially around the axial centerline; wherein the plurality of struts includes a first strut, a second strut and a third strut, and the first strut is oriented at a first clocking angle, wherein the first clocking angle is disposed between the major axis of the first strut and a first radial line that extends from the axial centerline and intersects the cross-section geometry center of the first strut, and the second strut is oriented at a second clocking angle, wherein the second clocking angle is disposed between the major axis of the second strut and a second radial line that extends from the axial centerline and intersects the cross-section geometry center of the second strut, wherein the second clocking angle is not equal to the first clocking angle, wherein the third strut is oriented at a third clocking angle, wherein the third clocking angle is disposed between the major axis of the third strut and a third radial line that extends from the axial centerline and intersects the cross-section geometry center of the third strut, and the third clocking angle is not equal to the first clocking angle or the second clocking angle; wherein the plurality of struts are non-uniformly circumferentially spaced within the air intake plenum; wherein the air intake plenum includes an air inlet; and wherein a majority of the struts within the plurality of struts are disposed within the air intake plenum in a portion of the air intake plenum disposed on an opposite side of the axial centerline than the air inlet.
2. A gas turbine engine, comprising: an axial centerline; at least one compressor rotor stage having a first bladed rotor rotatable about the axial centerline; at least one turbine rotor stage having a second bladed rotor rotatable about the axial centerline; and an air intake plenum in communication with the at least one compressor rotor stage, the air intake plenum including: a first gas path surface; a second gas path surface, spaced apart from the first gas path surface, wherein the first gas path surface and the second gas path surface define at least a portion of a plenum interior configured for air passage; and a plurality of struts that extend lengthwise between the first gas path surface and the second gas path surface through the plenum interior, each strut having a cross-section geometry within a plane perpendicular to a lengthwise axis of the strut, the cross-section geometry having a cross-section geometry center, a major axis, and a minor axis, where the major axis is greater than the minor axis, wherein the cross-section geometry is the same for each strut, and the cross-section geometry center disposed at an intersection of the major axis and the minor axis, and the cross-section is symmetrical about the cross-section geometry center; wherein the plurality of struts are disposed circumferentially spaced apart from one another within the air intake plenum, and the plurality of struts are disposed circumferentially around the axial centerline; and wherein the plurality of struts includes a first strut, a second strut and a third strut, and the first strut is oriented at a first clocking angle, wherein the first clocking angle is disposed between the major axis of the first strut and a first radial line that extends from the axial centerline and intersects the cross-section geometry center of the first strut, and the second strut is oriented at a second clocking angle, wherein the second clocking angle is disposed between the major axis of the second strut

and a second radial line that extends from the axial centerline and intersects the cross-section geometry center of the second strut, wherein the second clocking angle is not equal to the first clocking angle, wherein the third strut is oriented at a third clocking angle, wherein the third clocking angle is disposed between the major axis of the third strut and a third radial line that extends from the axial centerline and intersects the cross-section geometry center of the third strut, the third clocking angle is not equal to the first clocking angle or the second clocking angle; wherein the plurality of struts are non-uniformly circumferentially spaced within the air intake plenum; wherein the air intake plenum includes an air inlet; and wherein a majority of the struts within the plurality of struts are disposed within the air intake plenum in a portion of the air intake plenum disposed on an opposite side of the axial centerline than the air inlet.

3. An air intake plenum for a gas turbine engine, comprising: an axial centerline; an air inlet; a first gas path surface; a second gas path surface, spaced apart from the first gas path surface, wherein the first gas path surface and the second gas path surface define at least a portion of a plenum interior configured for air passage; and a plurality of struts that extend lengthwise between the first gas path surface and the second gas path surface through the plenum interior, each strut having a cross-section geometry within a plane perpendicular to a lengthwise axis of the strut, the cross-section geometry having a cross-section geometry center, a major axis, and a minor axis, where the major axis is greater than the minor axis; wherein the plurality of struts are disposed non-uniformly circumferentially spaced apart from one another within the air intake plenum, and a majority of the struts within the plurality of struts are disposed within the air intake plenum in a portion of the air intake plenum disposed on an opposite side of the axial centerline than the air inlet; and wherein the plurality of struts includes a first strut and a second strut, and the first strut is oriented at a first clocking angle, wherein the first clocking angle is disposed between the major axis of the first strut and a first radial line that extends from the axial centerline and intersects the cross-section geometry center of the first strut, and the second strut is oriented at a second clocking angle, wherein the second clocking angle is disposed between the major axis of the second strut and a second radial line that extends from the axial centerline and intersects the cross-section geometry center of the second strut, wherein the second clocking angle is not equal to the first clocking angle.

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